

Polar extremes

In the Arctic and Antarctic, the Sun disappears below the horizon for most of the winter. As air temperatures plunge far below the freezing point, the polar oceans freeze over, and a lot of the marine life moves away or lies dormant. But during the short summer, there is virtually continuous daylight, which melts the sea ice. Plankton multiplies explosively, causing a surge of breeding activity among polar animals before the seas freeze over again.



▲ ROSS SEA ICE

Dark water appears between drifting ice floes on the Ross Sea, Antarctica, as the ice breaks up in the weak summer sunshine.

FROZEN OCEANS

The Arctic Ocean is centered on the North Pole, but most of the Southern Ocean lies at some distance from the South Pole. This means that the central Arctic Ocean is colder than the Southern Ocean. As a result, the sea at the North Pole is permanently frozen over. But the vast mass of ice-covered rock at the South Pole is so intensely cold that it chills the air flowing off it, and this keeps the surface of the surrounding Southern Ocean frozen throughout the winter.

POLAR SUNLIGHT

The winter freeze is caused by the way the Sun never rises for long in the polar winter, so there is very little sunlight to warm the surface of the ocean. By contrast, the Sun never sets in the polar summer, but it is always very low in the sky because Earth's surface does not face toward the Sun in these regions. The Sun's rays are spread over a much wider area at the poles than they are near the equator, weakening their power and allowing some ice to survive at sea level throughout the summer.

